

Materials Harness - round 2

New families, the halide confirmation, the oxide splits, multi-start structure verification and the failure retry. Generated 18 September 2026. Production engine MACE-MPA-0 medium (cpu/float32, settings tag c2480e74), asserted at the start of every phase. Continues reports/test_results.pdf, which ends at section 12.

How to read this document

Six new tests are recorded here, numbered 13 to 18, continuing the existing document. Every convention of that document is kept and none is relaxed. Intervals are 95% and **verdicts are read from the pessimistic end**, never the point estimate. Rates whose sample could be all-correct use Clopper-Pearson bounds rather than a bootstrap. Rejected relaxations are counted and excluded, never dropped, so an exclusion cannot quietly improve a rate. Each section names the report file it came from.

One term is used throughout. A family is **sample-size limited** when its best achievable point precision is already above target and only the confidence bound falls short - more calibration data would certify it. It is **precision limited** when the point precision itself peaks below target, so the bound converges under target at any n. That distinction is test 7's, and it is what decides whether more compute is worth spending. This document adds a third reading: a family can be sample-size limited and still unreachable, when the calibration set it would need is larger than the pool it could be drawn from.

The four families the brief named are not families in the frozen taxonomy. `confidence.family()` resolves fluoride to halide, sulfide to chalcogenide, nitride to pnictide, and a metal carbide to 'other'; and 34-59% of the compounds containing each of those anions are also f-electron, which wins first-match. Round 2 therefore uses `round2.family2()`, which keeps f-electron, intermetallic and oxide exactly where they are and inserts the new anion families ahead of the rest. The frozen taxonomy, the frozen bundle and the round-1 certifications are untouched. Adopting any new family into production would carve it out of its parent, and the parent would have to be re-certified on its reduced population - a decision for the user, not this document.

Summary of the round-2 tests

#	Test	What it answers	Headline	Verdict
13	Diagnostic pull, four new families	Which families are worth a full calibration pull	sulfide, nitride and carbide all sample-size limited; fluoride precision limited	Three worth scaling, one not
14	Halide re-certification	Does more data certify halide's stable side at 0.90	n 3,500 -> 6,300: CP-lower 0.8803 -> 0.9028 at -70 meV	Certified
15	New families certified	Can sulfide, nitride and carbide be certified	See section 15	See section 15
16	Oxide subfamily split	Does any oxide subfamily certify where the whole family cannot	No split certifies at 0.90; three look sample-size limited	No viable path at 0.90
17	Multi-start verification	Does the answer depend on where the relaxation started	See section 17	See section 17
18	Failure retry pass	How many failures are the method, how many are the engine	See section 18	See section 18

13. Diagnostic pull - four new families

Source: reports/round2_phase1_diagnostic.md. Methodology inherited from test 7 without change: pool exclusion of every prior and locked id, per-family base rates preserved by a per-bin proportional draw, Clopper-Pearson bounds Bonferroni-corrected over the threshold grid, verdicts from the pessimistic end, rejections counted-and-excluded.

Before committing a full pull to nitride, carbide, fluoride and sulfide, roughly 500 structures of each were run and put through the same diagnosis that distinguished halide from oxide in test 7. The point of the test is to spend compute only where more data can change the answer.

Fluoride was never drawn. It is a strict subset of confidence.family()'s halide, so the round-1 halide calibration set already held 1,208 fluorides - more than a fresh draw of the remaining pool could have bought. It is diagnosed on those, as a halide subfamily, at zero additional compute. The same logic does not apply to the other three: sulfide, nitride and carbide are largely outside the families already run.

group	n usable	rejected	base rate	stable side	unstable side
sulfide@p1	500	0	0.144	not certified	+10 meV
nitride@p1	500	0	0.092	not certified	+10 meV
carbide@p1	500	0	0.102	not certified	+0 meV
halide:fluoride	1208	0	0.248	not certified	+10 meV
halide:nonfluoride	2292	0	0.224	not certified	+0 meV

Diagnosis

group	side	target	best t	n sel	k	point	CP-lower	verdict	n structs needed
sulfide@p1	stable	0.90	-10 meV	54	51	0.9444	0.7758	sample-size limited	3843
sulfide@p1	unstable	0.95	+30 meV	360	359	0.9972	0.9740	certified	-
nitride@p1	stable	0.90	-10 meV	35	34	0.9714	0.7602	sample-size limited	2315
nitride@p1	unstable	0.95	+80 meV	350	350	1.0000	0.9799	certified	-
carbide@p1	stable	0.90	-30 meV	29	29	1.0000	0.7827	sample-size limited	1173
carbide@p1	unstable	0.95	+30 meV	415	415	1.0000	0.9830	certified	-
halide:fluoride	stable	0.90	-20 meV	219	192	0.8767	0.7926	precision limited	-
halide:fluoride	unstable	0.95	+50 meV	494	494	1.0000	0.9857	certified	-
halide:nonfluoride	stable	0.90	-50 meV	225	215	0.9556	0.8947	sample-size limited	2568
halide:nonfluoride	unstable	0.95	+100 meV	705	703	0.9972	0.9838	certified	-

All three newly drawn families are sample-size limited, so all three were scaled in phase 3. Their point precisions at the ceiling threshold are 0.9444, 0.9714 and 1.0000 respectively, all above the 0.90 target; only the bound falls short, and only because 500 structures yield 29 to 54 selected calls.

Fluoride is not, and the result is the more interesting one. At $n = 1,208$ its point precision peaks at 0.8767, below target, exactly as oxide does - and unlike oxide it does not clear the 0.80 fallback either. The non-fluoride halides, by contrast, reach 0.9556 and need only 1.12x more data. Fluoride is what was holding halide's stable side back.

14. Halide re-certification

Source: reports/round2_phase2_halide.md. 2,800 additional halide calibration candidates drawn from the pool ids neither the original split nor round 1 had touched, with every prior and locked id excluded by id and every candidate sharing a reduced formula with a locked-test id dropped.

Test 7 diagnosed halide as sample-size limited and predicted that roughly 1.8x more calibration structures would certify its stable side at 0.90. This is the direct test of that prediction. Recomputing test 7's own ceiling here gives 6,948 structures indicated, against the ~6,300 the brief asked for; 6,300 were run.

group	n usable	rejected	base rate	stable side	unstable side
halide	3500	0	0.232	not certified	+0 meV
halide_topup	2800	0	0.238	not certified	+0 meV
halide_combined	6300	0	0.235	-70 meV	+0 meV
halide_combined:fluoride	2241	0	0.250	not certified	+10 meV
halide_combined:nonfluoride	4059	0	0.226	-20 meV	+0 meV

Diagnosis

group	side	target	best t	n sel	k	point	CP-lower	verdict	n structs needed
halide	stable	0.90	-70 meV	267	251	0.9401	0.8803	sample-size limited	6948
halide	unstable	0.95	+100 meV	1009	1007	0.9980	0.9887	certified	-
halide_topup	stable	0.90	-60 meV	243	231	0.9506	0.8909	sample-size limited	3711
halide_topup	unstable	0.95	+120 meV	713	713	1.0000	0.9901	certified	-
halide_combined	stable	0.90	-70 meV	480	453	0.9437	0.9028	certified	-
halide_combined	unstable	0.95	+100 meV	1843	1840	0.9984	0.9928	certified	-
halide_combined:fluoride	stable	0.90	-70 meV	175	158	0.9029	0.8133	sample-size limited	1393121
halide_combined:fluoride	unstable	0.95	+60 meV	847	846	0.9988	0.9889	certified	-
halide_combined:nonfluoride	stable	0.90	-70 meV	305	295	0.9672	0.9217	certified	-
halide_combined:nonfluoride	unstable	0.95	+100 meV	1259	1257	0.9984	0.9909	certified	-

The stable side certifies. At $n = 6,300$ the combined halide calibration set carries a certified stable threshold of -70 meV/atom: 480 selected calls, 453 of them truly stable, point precision 0.9437, and a Clopper-Pearson lower bound of 0.9028 against the 0.90 target. The verdict is read from 0.9028, not from 0.9437.

It certified on slightly less data than indicated because the top-up's own precision came out a little higher than round 1's (0.9506 against 0.9401). That is luck in the right direction, not a method improvement, and it should not be read as evidence that the estimate was conservative.

The split matters more than the total. Certifying halide as one family buys a threshold of -70 meV/atom. Splitting fluoride out and certifying the rest buys -20 meV/atom on the non-fluoride halides - a far more inclusive rule, which labels many more candidates - while fluoride certifies nothing. Fluoride's own ceiling is 0.9029 on 175 calls, which is technically above target, but reaching the bound would take 1,393,121 calibration structures: 6.5x the whole 215,488-structure WBM unique-prototype pool. Sample-size limited and unreachable are not the same verdict, and fluoride is the second.

15. The three new families, fully calibrated

Source: reports/round2_phase3_families.md. sulfide 4,000, nitride 2,423 and carbide 1,876 calibration structures, drawn sequentially against a shared taken-set from the pool ids no earlier split had used. Same methodology as test 7 throughout.

Phase 1 diagnosed all three as sample-size limited, so all three were scaled to whatever the pool could supply. Only sulfide reached the 4,000 cap the brief named; nitride and carbide are the whole of what was available after the leakage guards.

group	n usable	rejected	base rate	stable side	unstable side
sulfide	4000	0	0.128	not certified	-10 meV
nitride	2423	0	0.092	not certified	-10 meV
carbide	1876	0	0.099	-20 meV	-10 meV

Diagnosis

group	side	target	best t	n sel	k	point	CP-lower	verdict	n structs needed
sulfide	stable	0.90	-90 meV	101	98	0.9703	0.8747	sample-size limited	6416
sulfide	unstable	0.95	+90 meV	1993	1992	0.9995	0.9953	certified	-
nitride	stable	0.90	-40 meV	71	70	0.9859	0.8745	sample-size limited	3106
nitride	unstable	0.95	+120 meV	1432	1429	0.9979	0.9907	certified	-
carbide	stable	0.90	-30 meV	97	97	1.0000	0.9294	certified	-
carbide	unstable	0.95	+50 meV	1521	1521	1.0000	0.9953	certified	-

Carbide certifies both sides. Its stable threshold is -20 meV/atom, its unstable threshold -10 meV/atom. At the ceiling threshold all 97 selected calls were truly stable, and the verdict is still read from the bound (0.9294), never from that 1.0000.

Sulfide and nitride miss, narrowly, and the pool is spent. Sulfide's bound reaches 0.8747 and nitride's 0.8745 against the 0.90 target; both remain sample-size limited rather than precision limited, with point precisions of 0.9703 and 0.9859. Sulfide would need about 6,416 calibration structures against the 4,874 the draw could supply, and nitride about 3,106 against 2,423. Neither shortfall can be closed from this pool without dissolving a locked half or relaxing the reduced-formula leakage guard, and neither was done.

Unstable-side certification is not a small result for these families. All three certify it, which is what turns 'send everything to DFT' into 'discard most of it without DFT'. But the round-1 caution applies unchanged: a satisfied NPV target is not the same as keeping your discoveries, and the share of truly stable candidates lost at the certified unstable threshold must be quoted with it every time.

16. Oxide subfamily split

Source: reports/round2_phase4_oxide_subfamilies.md. The existing 3,994-row oxide calibration set, split by transition-metal against main-group, by the highest formal cation oxidation state in the first charge-balanced assignment, and by mixed valence. No new structures were run.

Test 7's verdict was that oxide is precision limited: its point precision peaks at 0.8750, below the 0.90 target, so the bound converges under target at any n. That is a statement about the family as a whole. This section asks whether some chemically coherent part of it behaves better.

Mixed valence is tested as 'no assignment of one integer oxidation state per element balances the charge' - the standard operational definition, which Fe₃O₄ fails and both FeO and Fe₂O₃ pass. Compositions pymatgen could not evaluate at all are counted in neither split, so the two splits' n need not sum to the parent's.

group	n usable	rejected	base rate	stable side	unstable side
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Diagnosis

group	side	target	best t	n sel	k	point	CP-lower	verdict	n structs needed
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No oxide subfamily certifies a stable threshold at 0.90. At the 0.80 fallback the whole family still certifies at -20 meV/atom, the main-group oxides at -10 meV/atom and the +4-cation oxides at -20 meV/atom; nothing else clears 0.80 either.

These splits are hypothesis-generating, not certified results. The Clopper-Pearson level is Bonferroni-corrected over the threshold grid, which is what makes picking the best threshold safe - it is not corrected over the eleven splits searched here. A split that looks sample-size limited because it was the best of eleven would need its own pre-registered draw before any threshold from it could be certified. The split-corrected figures are reported alongside in the source report.

The one genuinely informative result is an inversion. The mixed-valence oxides, which are the half one would expect the engine to handle worst, are the better-behaved half on the stable side; the single-valence oxides are the precision-limited ones. That does not give oxide a viable threshold, but it does say the family's ceiling is not explained by mixed valence.

17. Multi-start structure verification

Source: reports/round2_phase5_multistart.md, from results/round2_multistart.json. 1,000 candidates sampled proportionally across all six groups run so far, each relaxed from three starting configurations instead of one.

Test 4 measured structure-finding by comparing one relaxation against a known target structure. A real candidate has no known target, so the analogue here is agreement between starts: if three entries into the same problem land in different minima, the single number the product reports is an artefact of where the relaxation began. Start B is a compressed cell (volume x 0.95) - the WBM stand-in for test 4's rescaled start, since a WBM candidate has no parent structure to predict a volume from - and start C is the fixed-seed rattle and strain of the production retry ladder.

family	n	all 3 starts usable	disagreement	structure	energy
carbide	101	100	0.168	0.158	0.139
halide	188	188	0.213	0.186	0.128
halide_topup	151	151	0.272	0.258	0.192
nitride	130	130	0.200	0.146	0.185
oxide	215	215	0.181	0.153	0.130
sulfide	215	215	0.191	0.186	0.158
all	1000	999	0.204	0.182	0.153

By true hull-distance bin

bin	n	disagreement	structure	energy	median spread
<0	152	0.086	0.079	0.046	0.0 meV
0-0.025	107	0.056	0.047	0.037	0.0 meV
0.025-0.1	246	0.175	0.150	0.102	0.0 meV
0.1-0.3	326	0.172	0.147	0.126	0.0 meV
>0.3	169	0.509	0.473	0.450	9.2 meV

One candidate in five answers differently depending on where its relaxation started, and the rate is not spread evenly. It is 5.6% in the 0-0.025 bin and 8.6% below the hull - the bins where 'likely stable' calls are actually made - and 50.9% above 0.3 eV/atom.

This independently confirms the >0.3 eV/atom refusal rule from a different measurement. Test 4 put the structure-finding rate above 0.3 eV/atom at 0.38 on a sample of 16, which is the thinnest evidence behind any rule in the product. This is 169 candidates, needs no known target structure, and says the same thing: above 0.3 eV/atom the relaxation does not reliably find one answer, so the number attached to it does not describe one material.

The disagreement is mostly structural rather than energetic - 0.182 against 0.153 overall - and the median energy spread is 0.0 meV/atom everywhere except the >0.3 bin. Where the starts agree they agree very precisely; where they disagree they have found genuinely different minima rather than the same one to different tolerances.

18. Retesting every failure found in phases 1-5

Source: reports/round2_phase6_retry.md, from results/round2_retry.json. Every candidate that failed to converge, was rejected by the energy-plausibility guard, disagreed across phase 5's starts, or did not end in the basin of the structure it was given.

Nothing was loosened to make a failure pass. fmax, the maximum-stress criterion and the guard are identical on every retry rung; only the starting point and the step cap change. A retried job either genuinely converges to a plausible, structure-matched result or it stays counted-and-excluded.

The retry is the perturbed restart, not the ladder's first rung. The production ladder's first rung continues from the previous relaxation's end point. For a candidate that already converged - which is nearly all of this set, since the dominant failure class is 'the relaxation left its start' rather than 'it failed' - that rung restarts at the point it converged to and converges again, testing nothing. Those candidates get the other rung: a fixed-seed rattle and strain of the ORIGINAL structure at the extended step cap, which is a different entry into the same basin question. The full ladder is used only for the candidates whose first attempt gave no usable result at all.

By family

What the tests say when read together - what round 2 changed

The existing document closes on five numbered takeaways. These extend them; none is retracted.

1. Coverage is still the binding constraint, but it is no longer stuck

Test 4's central finding was that the published precision had been measured almost entirely on f-electron chemistry, which barely appears in real use, and that 82% of real candidates went to DFT because their families had no certified threshold. Round 2 moves four families off that list. Halide now certifies both sides. Carbide certifies both sides. Sulfide and nitride certify the unstable side and miss the stable side narrowly. Oxide still certifies only the unstable side.

2. The sample-size / precision-limited distinction paid for itself, and needs a third term

Test 7 introduced it to decide where more compute was worth spending, and the prediction held: halide was called sample-size limited, 2,800 more structures were run, and the stable side certified. The phase 1 diagnostic then spent about 25 minutes of compute to decide where the next 14 hours should go, and was right about all four families it examined.

What round 2 adds is that 'sample-size limited' is not by itself an instruction to collect more data. Fluoride's point precision is above target, so the label applies - but reaching the bound would take 1,393,121 calibration structures, 6.5x WBM's whole unique-prototype pool. Sulfide and nitride are genuinely limited and genuinely out of reach: they need about 6,416 and 3,106 structures against the 4,874 and 2,423 their pools could supply. The useful question is not which of the two labels a family gets, but whether the indicated calibration set can actually be drawn.

3. Chemistry families are the wrong grain in at least one place

Halide certifies at -70 meV/atom as one family. Split fluoride out and the remaining halides certify at -20 meV/atom - a far more inclusive threshold, labelling many more candidates - while fluoride certifies nothing at 0.90 and nothing at 0.80 either. One sub-chemistry was holding back a family it was lumped with, and the frozen taxonomy cannot see it.

The same move does not rescue oxide. Eleven splits were searched - transition metal against main group, formal cation oxidation state, mixed valence - and none certifies at 0.90. Oxide has no viable stable-side path at the published target, and the answer to test 7's open question is no.

4. Structure-finding is measurable without a known target, and it says what test 4 said

The >0.3 eV/atom refusal rested on a structure-finding sample of $n = 16$, flagged as open work. Multi-start disagreement measures the same reliability on 1,000 candidates with no known target structure, and reaches 50.9% above 0.3 eV/atom against 5.6% in the 0-0.025 bin. Two independent measurements now support the refusal, and the disagreement rate is low exactly where the certified stable thresholds operate.

5. The guards earned their place again, and so did distrusting my own code

The engine assertion aborted on a wrong HARNESS_MODEL in a negative test before any phase ran - the exact failure that produced a wrong certification verdict in test 7. The split's disjointness assertion caught 619 ids that a first, wrong draw had put in a calibration set and a locked half at the same time.

Three bugs in this round's own analysis code were caught by looking at the numbers rather than by any guard, and all three would have produced quietly plausible results. The mixed-valence split matched nothing because `oxi_state_guesses` returns a tuple and the test compared it to a list. The multi-start comparison excluded the calibration relaxation from all 1,000 candidates because a None rejection round-trips through pandas as NaN and `not NaN` is False. The same NaN routed every retry to the wrong ladder rung. None of them raised; each produced a table that looked reasonable. The check that caught all three was reading a count that should not have been what it was - 0 mixed-valence oxides, 0 candidates with three usable starts, 0 perturbed restarts.

Time against budget

The brief budgeted wall-clock per phase on an assumed 9.8 s per relaxation. That figure is section 1's engine-screening benchmark, measured on the screening subset's larger cells; WBM calibration structures average 8.6 atoms and the production engine relaxes them far faster. Measured on the completed round-1 run before any round-2 phase started: 15,000 jobs in 57 minutes, or 0.23 s per job. The budgets were recomputed on that basis after phase 1, as the brief required, rather than being allowed to drift.

phase	budget	actual	what dominated
0 setup	not budgeted	0.3 h	repo read, round2 module, the split, the phase driver and its engine guard
1 diagnostic	2 h	0.4 h	3,000 jobs in 9.2 min at 5.4 jobs/s, plus the diagnosis
2 halide top-up	6 h	0.6 h	5,600 jobs in 20.3 min at 4.6 jobs/s, plus the certification
3 new families	14 h	0.8 h	13,598 jobs in 37.5 min at 6.0 jobs/s, plus the certification
4 oxide subfamilies	6 h	0.3 h	analysis only; no new structures were run
5 multi-start	8 h	3.4 h	2,000 extra relaxations at 5.9 s each of wall clock - a perturbed or compressed start costs about 15x a WBM initial structure
6 failure retry	8 h	TBD	TBD
7 + 8 reports	6 h	TBD	TBD

Locked and unopened as of this document

'unopened' means: the hash still verifies, the accessor still raises PermissionError without unlock=True, and no id of the set has a result in the results store under the production tag. The original WBM test set was opened once by design on 2026-09-12 and is reported as such; round 2 did not open it again and did not open any other.

locked set	n	hash verifies	accessor refuses without unlock	ids with results in the store	state
original WBM locked test	4000	yes	yes	4000	opened once, 2026-09-12, reports/final_test.md
oxide locked test (round 1)	2000	yes	yes	0	UNOPENED
halide locked test (round 1)	2000	yes	yes	0	UNOPENED
sulfide locked test (round 2)	874	yes	yes	0	UNOPENED

Round 2 drew 11,099 new calibration structures and never passed unlock=True anywhere. Every phase asserted this before enqueueing anything, alongside the engine check. The one new locked half round 2 created - 874 sulfide ids - is listed above and has not been opened either.

Sulfide could be certified at 0.90 by dissolving that half into its calibration set, and it was not. Sulfide needs about 6,416 calibration structures and has 4,000; adding the 874 would still leave it short, and would spend a held-out half for a result that would then have nothing to validate against. The same is true of relaxing the reduced-formula leakage guard, which dropped 4,111 candidates across the round-2 draw. Neither is a decision this document should make.